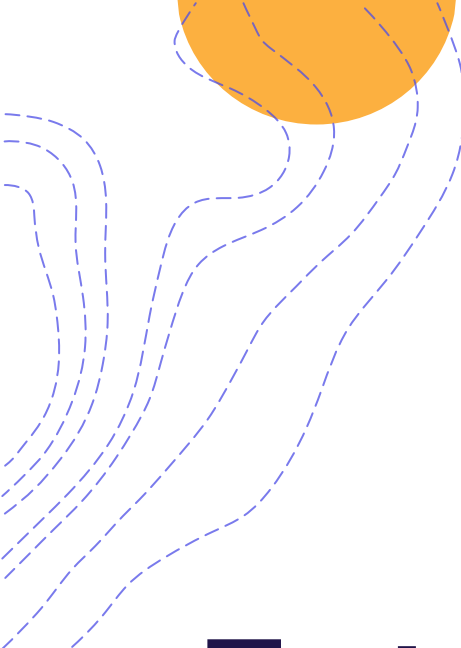




Future Directions of Adaptive Radiotherapy

From Adaptive Plans to Adaptive Care

Dandan Zheng, Laura Cerviño, Heng Li



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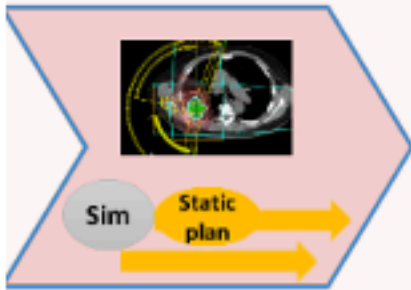
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Looking Back to Look Forward

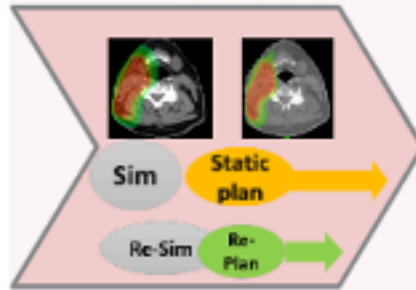
Adaptive Radiotherapy (ART) Development Timeline

Conventional RT



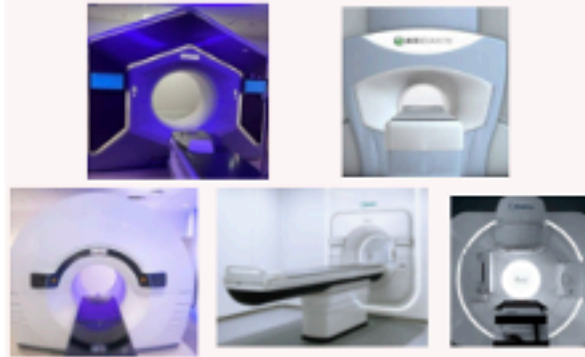
Single simulation;
static plan; no
adaptation

Offline ART



Anatomy variability
studies; mid-course
replanning

Online ART



Integrated platforms; daily
anatomy-based adjustments

Future Realtime ART



Ultra-fast workflows; AI-assisted
decisions; real-time adaptation



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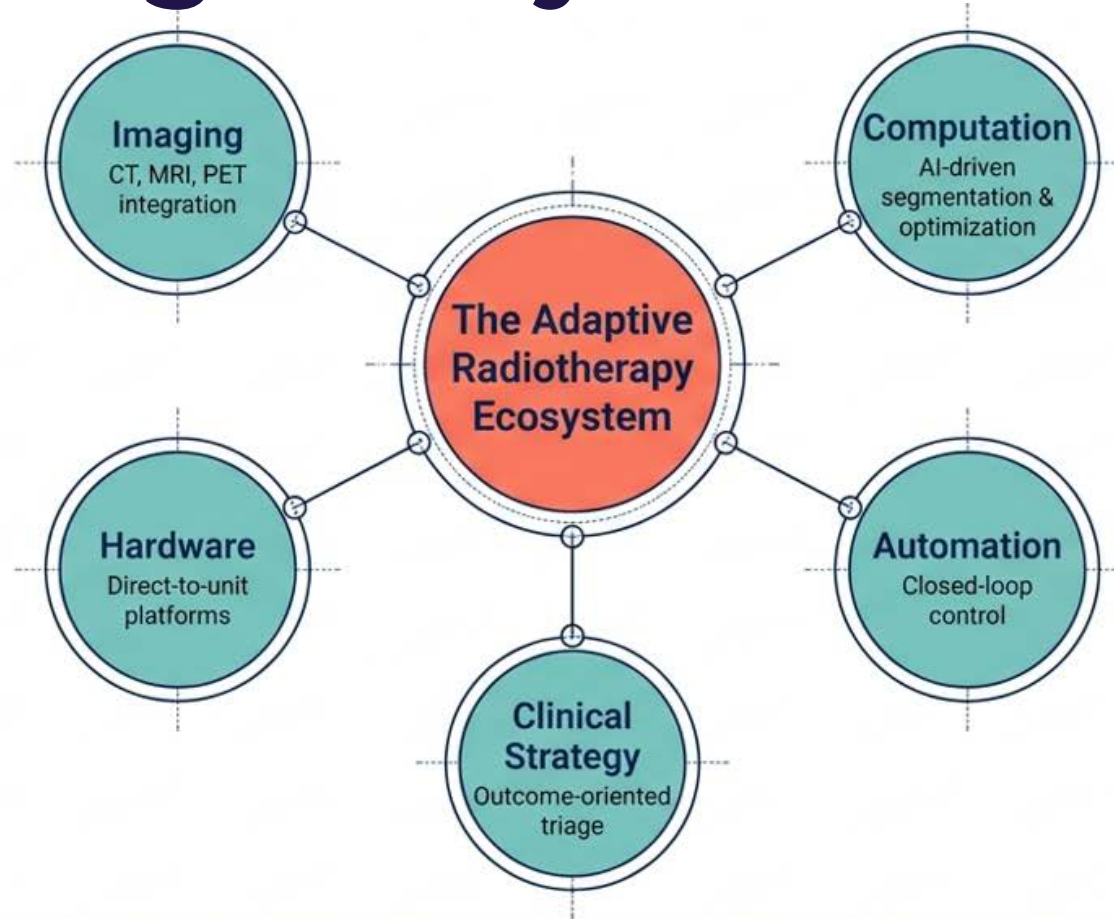


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The Evolving Ecosystem of ART



**Clinical evidence
accumulation**

The field is shifting from identifying **WHAT** can be adapted to determining **WHY, WHEN, and FOR WHOM** adaptation confers meaningful clinical benefit.

The Big Question

- When should we adapt?
- Why should we adapt?
- What outcome will it improve?

Two Possible Futures for ART

Dual Workflow Pathways

AI-Assisted Selective Adaptation

- Daily AI triage
- Adapt only when dosimetric benefit is predicted
- Expert oversight

Ultra-Fast Universal ART

- Routine daily adaptation
- Near-instant planning
- Minimal expert intervention

Future Integrated ART Platform



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Shrinking Timeline

Current State:
+15–30 Minute Delay

Future State:
< 3 Minutes

Auto-Segmentation

AI requiring zero
manual editing

Registration

Faster, fail-proof deformable
image registration

Optimization

Near-instant, high-quality
plan generation

Direct-to-Unit Platforms

High-quality volumetric imaging directly on the treatment unit eliminates the simulation step, anatomical mismatches, and time-to-treatment delays.

Clinical Impact: Particle Therapy Revival

Protons and heavy ions are hyper-sensitive to tissue density. Ultrafast ART ensures the Bragg peak is delivered accurately, restoring dosimetric advantages.



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What Must Improve?

Future Development Areas

Efficiency

- Increase automation & closed-loop control
- Reduce on-couch time
- Reduce expert workload

Robustness

- Increase accuracy
- Automated validation checks
- FMEA and guard against automation complacency

Applications

- Direct-to-unit treatments
- Proton & particle therapy
- Biology-guided ART



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Considerations

- The Direct-to-Unit Future: Image on treatment unit → Plan → Treat
- Why Adaptive Proton Therapy Matters: Bragg peak and sensitivity to anatomy.
- Which Patients Actually Need ART? Smarter ART, Not More ART
- Real-Time ART: Image → Predict → Adapt → Deliver → Repeat
- Beyond Anatomy:
 - Today: Anatomy-guided ART
 - Future: Biology-guided ART



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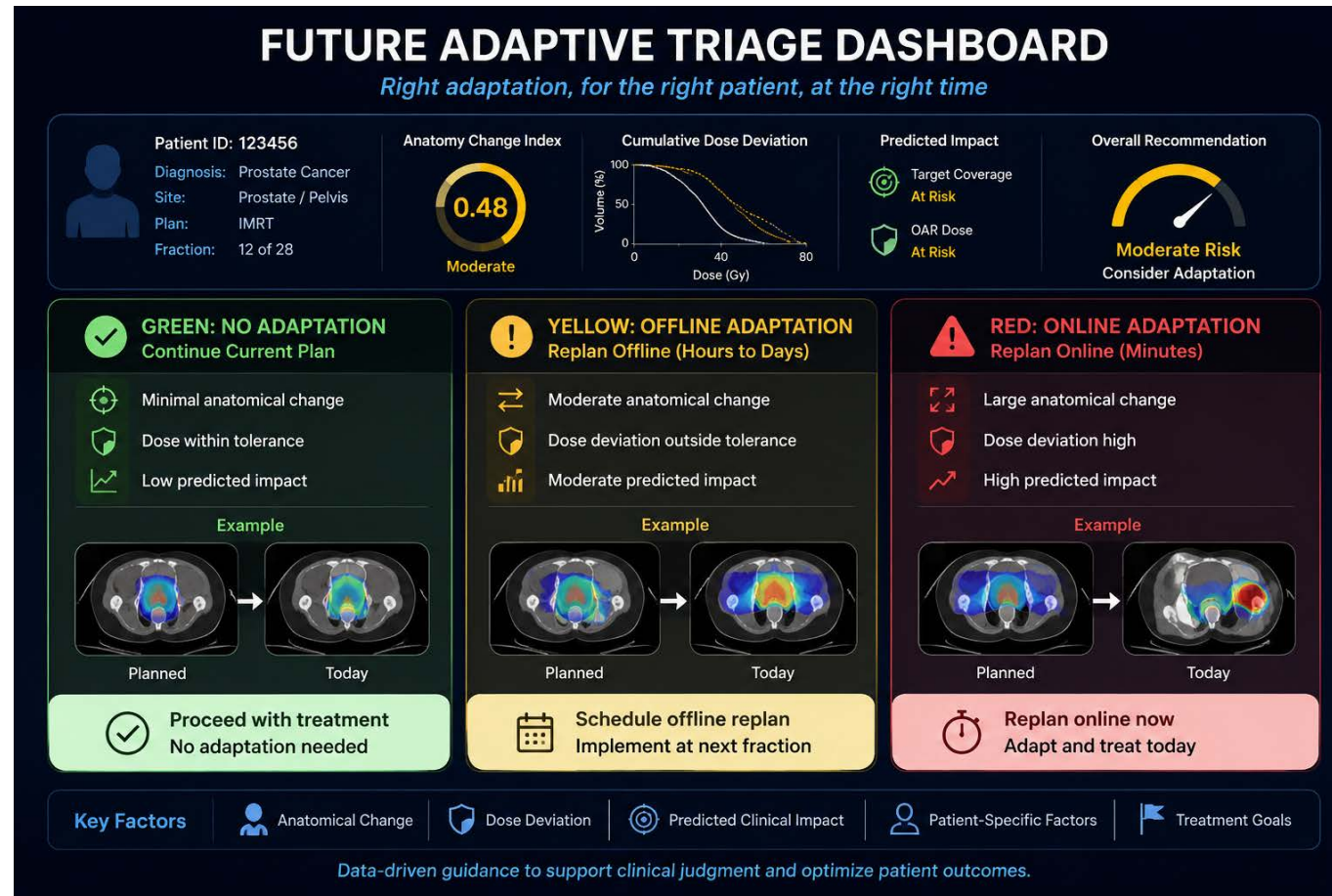


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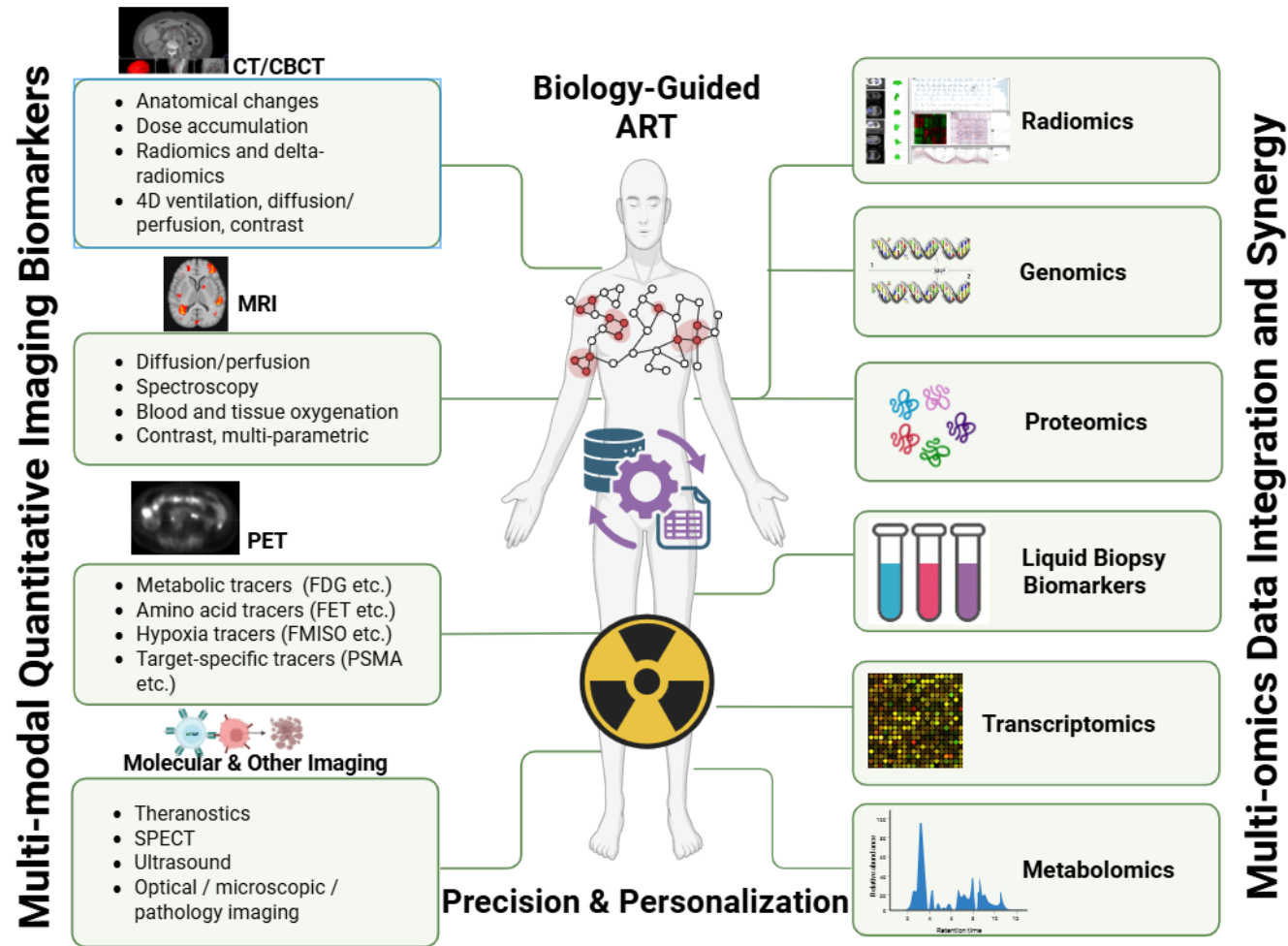
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Future Adaptive Triage Dashboard



Biology-Guided ART



Anatomical Volatility (Low to High)
(e.g., Mucosa changes, edema, organ filling)



Biological Evolution (Low to High)
(e.g., Hypoxia evolution, early response)



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Other Considerations

- Ethics, Accessibility, and AI: Bias, Access, Privacy
- Outcome-Oriented ART: Dosimetry → Clinical Benefit → Patient Benefit → Value
- Learning Health System: Patients → Data → Outcomes → Better Adaptation



What do you think will have the biggest impact on ART over the next 10 years?

Closing

- The right treatment, for the right patient, at the right time.